

Venkat Sai Jinka

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Summary:

Detail-oriented and innovative Senior Data Scientist with over 5+ years of experience specializing in Generative AI, Machine Learning (ML), Deep Learning (DL), and Natural Language Processing (NLP). Expertise in designing, developing, and deploying AI models for applications such as content creation, data augmentation, and anomaly detection. Proven track record in implementing Generative AI systems, large language models (LLMs), and advanced NLP techniques for driving business value.

Technical Skills:

- Programming Languages: Python, R, SQL
- Data Analysis & Libraries: Pandas, NumPy, Matplotlib, SciPy, Scikit-learn
- Deep Learning Frameworks: TensorFlow, PyTorch, Keras
- Natural Language Processing: NLTK, spaCy, BERT, GPT, Word2Vec, GloVe, FastText, ELMo, OpenAI-Embeddings
- Generative AI: OpenAI LLMs, Hugging Face Models, GANs, VAEs, Transformers, LangChain
- Vector Databases: Chroma, Faiss, Milvus, Pinecone
- Cloud Platforms: AWS, Azure, Google Cloud
- Data Visualization: Power BI, Tableau, Plotly
- Version Control: Git, Bitbucket, TFS

Professional Experience:

Senior Data Scientist | Bank of America | Remote | Aug 2024 – Present

- Led the development of an automated report generation system using Generative AI (GenAI) and LLMs (Llama 3)
- Implemented Retrieval-Augmented Generation (RAG) framework using knowledge graphs and vector databases (Neo4j, FAISS), improving contextual retrieval accuracy by 35%.
- Engineered data pipelines using Pandas and NumPy, streamlining preprocessing for ML models.
- Integrated OCR solutions (Tesseract, AWS Textract) to automate document digitization, improving efficiency by 40%.
- Designed tailored prompts for LLMs to generate human-quality text sections for reports.

- Developed and deployed autonomous AI agents using Microsoft's Autogen for document summarization, financial analysis, and fact-checking.
- Improved operational efficiency by 40% through multi-agent orchestration in enterprise systems.
- Built and deployed machine learning models using AWS SageMaker, optimizing credit risk assessment and customer segmentation.

Data Scientist | OPTUM, UnitedHealthcare | Hyderabad, India | Jan 2022

– Aug 2023

- Developed ML models for fraud detection using scikit-learn and deep learning with TensorFlow.
- Built interactive dashboards in Tableau and Power BI, enabling real-time business insights and improved decision-making.
- Leveraged NLP for information retrieval and sentiment analysis, increasing retrieval performance by 30%.
- Implemented advanced ensemble learning methods (stacking, gradient boosting) to improve fraud detection performance.
- Deployed real-time scoring pipelines using Flask, enabling seamless integration with healthcare systems.

Associate Data Scientist | Elevance Health | Hyderabad, India | May 2018

– Dec 2021

- Performed data preprocessing, feature engineering, and EDA on customer data for predictive modeling.
- Applied clustering algorithms (k-means, hierarchical) for customer segmentation and developed predictive models using logistic regression and random forests.
- Built and optimized ETL pipelines with AWS Glue, improving data processing time by 30%.
- Developed SQL queries for efficient data retrieval from relational databases, enhancing ML model performance.

Education:

- Master's in Business Analytics, Trine University, 2025
- MSCS, AIISR, 2019